Volume & 3D Shapes

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 12cm³

2. 6

3. B — cube

4. 12

5. 8

6. 8cm³

7. C — sphere

8. 8cm³

9. A, C and E (a sphere, cylinder and cone all have a curved surface)

10. 6

11. 15cm³

12. B — a cube is a special type of cuboid (all sides equal)

13. 12

14. 6cm³

15. 8

Standard

16. 24cm³

17. C — both A (24cm³) and B (24cm³)

18. 20cm³

19. 5

20. 6

21. B — cylinder

22. 2cm

23. A and C (volumes 27cm³ and 30cm³ are more than 20cm³; B=16cm³, D=12cm³ and E=8cm³ are not)

24. 8

25. 5

26. 36cm³

27. B — square-based pyramid

28. 27cm³

29. 2

30. 24cm³

Standard / Hard

31. 24 ÷ (4×3) = 2cm

32. A, B and D (the box holds 24 unit cubes, its volume is 24cm³, and each small cube is 1cm³)

33. Neither — both have a volume of 24cm³ (4×2×3 = 3×4×2 = 24)

34. 40 ÷ (5×4) = 2cm

35. A — 2cm (since 2×2×2=8)

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36. Dimensions 4cm × 1cm × 1cm; volume = 4×1×1 = 4cm³

37. C — 8×5×2 = 80cm³

38. A — 4×3×2 = 24cm³, the same as 6×2×2 = 24cm³; B gives 3×4×1 = 12cm³, which is different

39. Side = 3cm (since 3×3=9); volume = 3×3×3 = 27cm³

40. No — a cube needs all 3 sides equal, and here two sides are 4cm but the third is 2cm; volume = 4×4×2 = 32cm³

41. 3×3×4 = 36cm³

42. 8×2×3 = 48cm³

43. A, B, C and E (all give a volume of 12cm³; D gives 16cm³)

44. Total cubes = 5×2×2 = 20; half removed leaves 10 cubes

45. 5×3×2 = 30cm³

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Homework Worksheet — Answers

Easy / Standard

1. 16cm³

2. 6

3. B — triangular prism

4. 12

5. 8

6. 10cm³

7. C — cone

8. 27cm³

9. B, C, D and E (all except the sphere have at least one flat face — the cylinder and cone each have flat circular faces)

10. 6

11. 12cm³

12. B — cube

13. 12

14. 7cm³

15. 8

Standard

16. 18cm³

17. C — both A (20cm³) and B (20cm³)

18. 12cm³

19. 5

20. 1

21. B — triangular prism

22. 3cm

23. A, B and E (volumes 16cm³, 18cm³ and 16cm³ are all more than 15cm³; C=10, D=8 are not)

24. 9

25. 3

26. 16cm³

27. B — cube

28. 64cm³

29. 6

30. 30cm³

Standard / Advanced

31. Height = 2×2 = 4cm; volume = 3×2×4 = 24cm³

32. B — edge = 3cm, and 3×3×3 = 27cm³

33. Cube's volume = 2×2×2 = 8cm³, so height = 8 ÷ (4×1) = 2cm

34. Cuboid's volume = 4×2×1 = 8cm³, so the cube's edge = 2cm (since 2×2×2=8)

35. Tank's volume = 8×5×2 = 80cm³; half full = 40cm³

36. Edge = 4cm (since 4×4×4=64); 6 faces

37. A — 48 ÷ (6×4) = 2cm

38. Each cube's volume = 2×2×2 = 8cm³; total = 3×8 = 24cm³; height = 3×2 = 6cm

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39. A, B, C and E (all give a volume of 24cm³; D gives 16cm³)

40. 6cm × 4cm × 1cm each

41. Width × length = 40 ÷ 5 = 8, and length = 2 × width, so width = 2cm, length = 4cm (check: 4×2×5 = 40)

42. Container's volume = 8×5×3 = 120cm³; 120 ÷ 20 = 6 bottles exactly, with 0cm³ left over

43. Tank's volume = 10×5×4 = 200cm³; half full = 100cm³; plus 50cm³ = 150cm³

44. 4×3×1 = 12cm³

45. C — new height = 2×2 = 4cm; new volume = 6×4×4 = 96cm³

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